Question 1:What type of cable is used to connect the Ethernet interface on a host PC to the Ethernet interface on a switch?
Answer: Copper Straight-Through

Question 2: What type of cable is used to connect the Ethernet interface on a switch to the Ethernet interface on a router?
Answer: Copper Straight-Through

Question 3: What type of cable is used to connect the Ethernet interface on a router to the Ethernet interface on a host PC?
Answer: Copper Cross-Over

Question 4: What would happen if you answered yes to the question, “System configuration has been modified. Save?”
Answer: Answering "yes" to save the modified configuration would overwrite the startup configuration in NVRAM, retaining the old settings and defeating the purpose of erasing the configuration to start clean before the reload.

Question 5: Why would you want to disable DNS lookup in a lab environment?
Answer: To prevent the router from trying to resolve unrecognized commands as hostnames, which can cause delays in the command-line interface (CLI).

Question 6: What would happen if you disabled DNS lookup in a production environment?
Answer: Disabling DNS lookup in a production environment means the router won't be able to resolve domain names to IP addresses. This could affect network operations that rely on domain name resolution.

Question 7: Why is it not necessary to use the enable password password command?
Answer: It's not necessary to use the enable password command because the enable secret command is superior. It offers better security by encrypting the password, and crucially it takes precedence over and overrides the unencrypted enable password when both are configured.

Question 8: When does this banner display?
Answer: The banner displays immediately upon accessing the router.

Question 9: Why should every router have a message-of-the-day banner?
Answer: To warn and deter unauthorized access, whether intentional or accidental, and to provide legal notice that the system is restricted. It can also be used to provide a message or any guideline for users accessing the device.

Question 10: What is a shorter version of this command?
Answer: copy run start

Question 11: From the host attached to R1, is it possible to ping the default gateway?
Answer: Yes

Question 12: From the host attached to R2, is it possible to ping the default gateway?
Answer: Yes

Question 13: Check the PCs. Are they physically connected to the correct router? (Connection could be through a switch or directly.)
Answer: Yes

Question 14: Are link lights blinking on all relevant ports?
Answer: Yes

Question 15: Check the PC configurations. Do they match the Topology Diagram?
Answer: Yes

Question 16: Check the router interfaces using the show ip interface brief command. Are the interfaces “up” and “up”?
Answer: Yes

Question 17: From the router R1, is it possible to ping R2 using the command ping 192.168.2.2?
Answer: Yes

Question 18: From the router R2, is it possible to ping R1 using the command ping 192.168.2.1?
Answer: Yes

Question 19: Check the cabling. Are the routers physically connected?
Answer: Yes

Question 20: Are link lights blinking on all relevant ports?
Answer: Yes

Question 21:Check the router configurations. Do they match the Topology Diagram?
Answer: Yes

Question 22: Did you configure the clock rate command on the DCE side of the link?
Answer: Yes

Question 23: Check the router interfaces using the show ip interface brief command. Are the interfaces “up” and “up”?
Answer: Yes

Question 24: What is missing from the network that is preventing communication between these devices?
Answer: The network is missing a dynamic routing protocol or static routes, which are necessary for the routers (R1 and R2) to know how to reach networks they are not directly connected to.

For R1:

show ip route

```
R1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial0/0/0
```

show ip interface brief

```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet0/1        unassigned      YES NVRAM  administratively down down
Serial0/0/0            192.168.2.1     YES manual up                    up
Serial0/0/1            unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

For R2:

show ip route

```
R1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial0/0/0
```

show ip interface brief

```
R2>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.3.1     YES manual up                    up
FastEthernet0/1        unassigned      YES NVRAM  administratively down down
Serial0/0/0            192.168.2.2     YES manual up                    up
Serial0/0/1            unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```